# Output Before Data Cleaning:

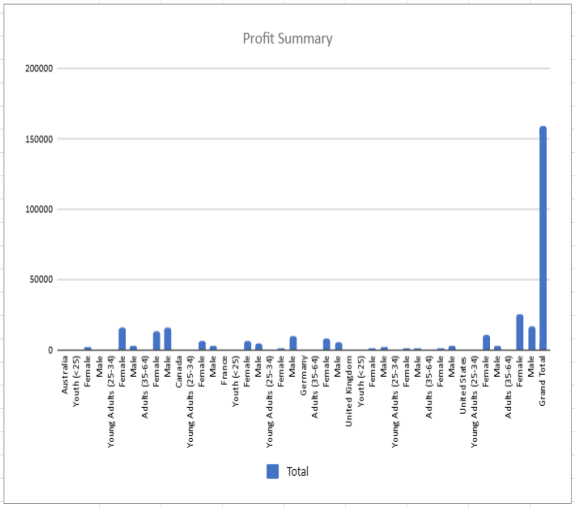
| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Sales_Order # | Date | Day | Month | Year | Customer | Age_Group | Customer | Country | State | Product_( | Sub_Category | Product_Description | Order_Qu | Unit_Cost | Unit_Price | Profit |
| 2 | 000261695 | 12/1/2021 | 1 | December | 2021 | 39 | Adults (35-64) | F | United States | California | Bikes | Mountain Bikes | Mountain-200 Black, 46 | 4 | $1,252.00 | $2,295.00 | $4,172 |
| 3 | 000261695 | 12/1/2021 | 1 | December | 2021 | 44 | Adults (35-64) | M | United Kingdom | England | Bikes | Mountain Bikes | Mountain-200 Silver, 42 | 1 | $1,266.00 | $2,320.00 | $1,054 |
| 4 | 000261697 | 12/2/2021 | 2 | December | 2021 | 37 | Adults (35-64) | M | United States | California | Bikes | Mountain Bikes | Mountain-400-W Silver, 46 | 2 | $420.00 | $769.00 | $698 |
| 5 | 000261698 | 12/2/2021 | 2 | December | 2021 | 31 | Young Adults (25-34) | F | Australia | New South Wales | Bikes | Mountain Bikes | Mountain-400-W Silver, 42 | 1 | $420.00 | $769.00 | $349 |
| 6 | 000261699 | 12/3/2021 | 3 | December | 2021 | 37 | Adults (35-64) | F | United States | California | Bikes | Mountain Bikes | Mountain-200 Black, 46 | 2 | $0.00 | $2,295.00 | $2,086 |
| 7 | 000261700 | 12/3/2021 | 3 | December | 2021 | 24 | Youth (<25) | F | United Kingdom | England | Bikes | Mountain Bikes | Mountain-200 Black, 38 | 1 | $1,252.00 | $2,295.00 | $1,043 |
| 8 | 000261701 | 12/3/2021 | 3 | December | 2021 | 37 | Adults (35-64) | M | United States | Washington | Bikes | Mountain Bikes | Mountain-200 Black, 46 | 1 | $1,252.00 | $2,295.00 | $1,043 |
| 9 | 000261701 | 12/3/2021 | 3 | December | 2021 | 37 | Adults (35-64) | M | United States | Washington | Bikes | Mountain Bikes | Mountain-200 Black, 46 | 1 | $1,252.00 | $2,295.00 | $1,043 |
| 10 | 000261702 | 12/4/2021 | 4 | December | 2021 | 31 | Young Adults (25-34) | F | Australia | New South Wales | Bikes | Mountain Bikes | Mountain-400-W Silver, 42 | 4 | $420.00 | $0.00 | $1,396 |
| 11 | 000261703 | 12/5/2021 | 5 | December | 2021 | 39 | Adults (35-64) | F | United States | California | Bikes | Mountain Bikes | Mountain-200 Black, 46 | 4 | $1,252.00 | $2,295.00 | $4,172 |
| 12 | 000261704 | 12/5/2021 | | December | 2021 | 42 | Adults (35-64) | M | Germany | Nordrhein-Westfalen | Bikes | Mountain Bikes | Mountain-200 Black, 38 | 4 | $1,252.00 | $2,295.00 | $4,172 |
| 13 | 000261705 | 12/5/2021 | 5 | December | 2021 | 35 | Adults (35-64) | F | Australia | Queensland | Bikes | Mountain Bikes | Mountain-200 Silver, 38 | 1 | $1,266.00 | $2,320.00 | $1,054 |
| 14 | 000261706 | 12/5/2021 | 5 | December | 2021 | 37 | Adults (35-64) | F | United States | California | Bikes | Mountain Bikes | Mountain-200 Black, 46 | 1 | $1,252.00 | $2,295.00 | $1,04 |
| 15 | 000261707 | 12/6/2021 | 6 | December | 2021 | 23 | Youth (<25) | M | United Kingdom | England | Bikes | Mountain Bikes | Mountain-400-W Silver, 46 | 3 | $420.00 | $769.00 | $1,04 |
| 16 | 000261708 | 12/6/2021 | 6 | December | 2021 | 27 | Young Adults (25-34) | M | Canada | British Columbia | Bikes | Mountain Bikes | Mountain-200 Black, 46 | 1 | $1,252.00 | $2,295.00 | $1,04 |
| 17 | 000261709 | 12/6/2021 | 6 | December | 2021 | 36 | | M | Australia | New South Wales | Bikes | Mountain Bikes | Mountain-200 Black, 42 | 1 | $1,252.00 | $2,295.00 | $1,04 |
| 18 | 000261710 | 12/6/2021 | 6 | December | 2021 | 47 | Adults (35-64) | M | United Kingdom | England | Bikes | Mountain Bikes | Mountain-200 Silver, 38 | 1 | $1,266.00 | $2,320.00 | $1,054 |
| 19 | 000261711 | 12/7/2021 | 7 | Decmber | 2021 | 30 | Young Adults (25-34) | M | United States | California | Bikes | Mountain Bikes | Mountain-400-W Silver, 38 | 4 | $420.00 | $769.00 | $1,396 |
| 20 | 000261712 | 12/7/2021 | 7 | December | 2021 | 38 | Adults (35-64) | M | United States | California | Bikes | Mountain Bikes | Mountain-200 Silver, 42 | 2 | $1,266.00 | $2,320.00 | $2,108 |
| 21 | 000261713 | 12/8/2021 | 8 | December | 2021 | 19 | Youth (<25) | F | Australia | New South Wales | Bikes | Mountain Bikes | Mountain-500 Silver, 42 | 4 | $308.00 | $565.00 | $1,028 |
| 22 | 000261714 | 12/8/2021 | 8 | December | 2021 | 30 | Young Adults (25-34) | F | Canada | British Columbia | Bikes | Mountain Bikes | Mountain-200 Silver, 38 | 4 | $1,266.00 | $2,320.00 | $4,216 |
| 23 | 000261715 | 12/8/2021 | 8 | December | 2021 | 39 | Adults (35-64) | F | United States | Oregon | Bikes | Mountain Bikes | | 2 | $1,252.00 | $2,295.00 | $2,086 |
| 24 | 000261716 | 12/8/2021 | 8 | December | 2021 | 35 | Adults (35-64) | F | United States | California | Bikes | Mountain Bikes | Mountain-500 Black, 42 | | $295.00 | $540.00 | $24 |
| 25 | 000261717 | 12/9/2021 | 9 | December | 2021 | 33 | Young Adults (25-34) | F | Australia | Victoria | Bikes | Mountain Bikes | Mountain-100 Black, 38 | 2 | $1,898.00 | $3,375.00 | $2,954 |
| 26 | 000261718 | 12/9/2021 | 9 | December | 2021 | 41 | Adults (35-64) | F | Germany | Hamburg | Bikes | Mountain Bikes | Mountain-200 Silver, 42 | 1 | $1,266.00 | $2,320.00 | $1,054 |
| 27 | 000261719 | 12/10/2021 | 10 | December | 2021 | 34 | Young Adults (25-34) | F | United States | California | Bikes | Mountain Bikes | Mountain-200 Black, 42 | 2 | $1,252.00 | $2,295.00 | $2,086 |

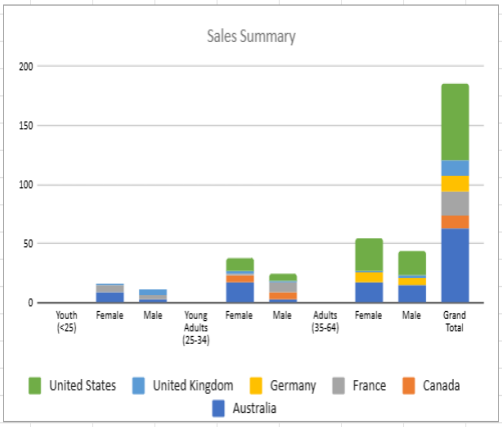
# Output After Data Cleaning:

| | A | B | C | D | E | F | G | H | I | J | K | L | M |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Sales_Order # | Date | Day | Month | Year | Customer | Age_Group | Customer_Gender | Country | State | Product Category | Subcategory | Model |
| 2 | 000261695 | 12/1/2021 | 1 | December | 2021 | 39 | Adults (35-64) | Female | United States | California | Bikes | Mountain Bikes | Mountain-200 |
| 3 | 000261696 | 12/1/2021 | 1 | December | 2021 | 44 | Adults (35-64) | Male | United Kingdom | England | Bikes | Mountain Bikes | Mountain-200 |
| 4 | 000261697 | 12/2/2021 | 2 | December | 2021 | 37 | Adults (35-64) | Male | United States | California | Bikes | Mountain Bikes | Mountain-400-W |
| 5 | 000261698 | 12/2/2021 | 2 | December | 2021 | 31 | Young Adults (25-34) | Female | Australia | New South Wales | Bikes | Mountain Bikes | Mountain-400-W |
| 6 | 000261699 | 12/3/2021 | 3 | December | 2021 | 37 | Adults (35-64) | Female | United States | California | Bikes | Mountain Bikes | Mountain-200 |
| 7 | 000261700 | 12/3/2021 | 3 | December | 2021 | 24 | Youth (<25) | Female | United Kingdom | England | Bikes | Mountain Bikes | Mountain-200 |
| 8 | 000261701 | 12/3/2021 | 3 | December | 2021 | 37 | Adults (35-64) | Male | United States | Washington | Bikes | Mountain Bikes | Mountain-200 |
| 9 | 000261702 | 12/4/2021 | 4 | December | 2021 | 31 | Young Adults (25-34) | Female | Australia | New South Wales | Bikes | Mountain Bikes | Mountain-400-W |
| 10 | 000261703 | 12/5/2021 | 5 | December | 2021 | 39 | Adults (35-64) | Female | United States | California | Bikes | Mountain Bikes | Mountain-200 |
| 11 | 000261704 | 12/5/2021 | 5 | December | 2021 | 42 | Adults (35-64) | Male | Germany | Nordrhein-Westfalen | Bikes | Mountain Bikes | Mountain-200 |
| 12 | 000261705 | 12/5/2021 | 5 | December | 2021 | 35 | Adults (35-64) | Female | Australia | Queensland | Bikes | Mountain Bikes | Mountain-200 |
| 13 | 000261706 | 12/5/2021 | 5 | December | 2021 | 37 | Adults (35-64) | Female | United States | California | Bikes | Mountain Bikes | Mountain-200 |
| 14 | 000261707 | 12/6/2021 | 6 | December | 2021 | 23 | Youth (<25) | Male | United Kingdom | England | Bikes | Mountain Bikes | Mountain-400-W |
| 15 | 000261708 | 12/6/2021 | 6 | December | 2021 | 27 | Young Adults (25-34) | Male | Canada | British Columbia | Bikes | Mountain Bikes | Mountain-200 |
| 16 | 000261709 | 12/6/2021 | 6 | December | 2021 | 36 | Youth (<25) | Male | Australia | New South Wales | Bikes | Mountain Bikes | Mountain-200 |
| 17 | 000261710 | 12/6/2021 | 6 | December | 2021 | 47 | Adults (35-64) | Male | United Kingdom | England | Bikes | Mountain Bikes | Mountain-200 |
| 18 | 000261711 | 12/7/2021 | 7 | December | 2021 | 30 | Young Adults (25-34) | Male | United States | California | Bikes | Mountain Bikes | Mountain-400-W |
| 19 | 000261712 | 12/7/2021 | 7 | December | 2021 | 38 | Adults (35-64) | Male | United States | California | Bikes | Mountain Bikes | Mountain-200 |
| 20 | 000261713 | 12/8/2021 | 8 | December | 2021 | 19 | Youth (<25) | Female | Australia | New South Wales | Bikes | Mountain Bikes | Mountain-500 |
| 21 | 000261714 | 12/8/2021 | 8 | December | 2021 | 30 | Young Adults (25-34) | Female | Canada | British Columbia | Bikes | Mountain Bikes | Mountain-200 |
| 22 | 000261715 | 12/8/2021 | 8 | December | 2021 | 39 | Adults (35-64) | Female | United States | Oregon | Bikes | Mountain Bikes | Mountain-200 |
| 23 | 000261716 | 12/8/2021 | 8 | December | 2021 | 35 | Adults (35-64) | Female | United States | California | Bikes | Mountain Bikes | Mountain-500 |
| 24 | 000261717 | 12/9/2021 | 9 | December | 2021 | 33 | Young Adults (25-34) | Female | Australia | Victoria | Bikes | Mountain Bikes | Mountain-100 |
| 25 | 000261718 | 12/9/2021 | 9 | December | 2021 | 41 | Adults (35-64) | Female | Germany | Hamburg | Bikes | Mountain Bikes | Mountain-200 |
| 26 | 000261719 | 12/10/2021 | 10 | December | 2021 | 34 | Young Adults (25-34) | Female | United States | California | Bikes | Mountain Bikes | Mountain-200 |
| 27 | 000261720 | 12/10/2021 | 10 | December | 2021 | 40 | Adults (35-64) | Male | Australia | New South Wales | Bikes | Mountain Bikes | Mountain-200 |

# Challenge Activity Profit:

| Age_Group | Sum of Profit |
|---|---|
| Australia | |
| Youth (<25) | |
| Female | 2265 |
| Male | 490 |
| Young Adults (25-34) | |
| Female | 16204 |
| Male | 2435 |
| Adults (35-64) | |
| Female | 13276 |
| Male | 15656 |
| Canada | |
| Young Adults (25-34) | |
| Female | 6302 |
| Male | 2821 |
| France | |
| Youth (<25) | |
| Female | 6291 |
| Male | 4216 |
| Young Adults (25-34) | |
| Female | 1054 |
| Male | 9420 |
| Germany | |
| Adults (35-64) | |
| Female | 8410 |
| Male | 5226 |




# Data Analysis:

| Sum of Order_Quantity | COUNTRY | | | | | |
|---|---|---|---|---|---|---|
| Age_Group | Australia | Canada | France | Germany | United Kingdom | United States |
| Youth (<25) | | | | | | |
| Female | 9 | 0 | 6 | 0 | 1 | 0 |
| Male | 2 | 0 | 4 | 0 | 5 | 0 |
| Young Adults (25-34) | | | | | | |
| Female | 17 | 6 | 1 | 0 | 3 | 10 |
| Male | 3 | 5 | 9 | 0 | 1 | 6 |
| Adults (35-64) | | | | | | |
| Female | 17 | 0 | 0 | 8 | 1 | 28 |
| Male | 15 | 0 | 0 | 5 | 3 | 21 |
| Grand Total | 63 | 11 | 20 | 13 | 14 | 65 |